

Brutal Practice Paper B10 - Nutrition in Animals, Simple Equations, Electric Current, Fractions & Decimals

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A cow's stomach has four chambers. Grass spends 30 minutes in the rumen, then twice that long in the reticulum, then 45 minutes in the omasum. If the abomasum takes as long as the first three chambers put together, how long is the whole process? [\[Nutrition in Animals\]](#)
 - (A) 165 minutes
 - (B) 90 minutes
 - (C) 405 minutes
 - (D) 135 minutes
 - (E) 270 minutes
 - (F) 210 minutes

2. Solve for x: $3(2x - 5) + 4 = 2(x + 7) - 5$. [\[Simple Equations\]](#)
 - (A) 4
 - (B) 5
 - (C) 3
 - (D) 2
 - (E) 7
 - (F) 6

3. A torch holds 3 cells in series, each 1.5 V, lighting a bulb that needs 4.5 V to glow fully. One cell goes completely dead (0 V). What voltage now reaches the bulb, and does it glow fully? [\[Electric Current & its Effects\]](#)
 - (A) 1.5 V, does not glow fully
 - (B) 6.0 V, glows fully
 - (C) 3.0 V, so it does not glow fully
 - (D) 4.5 V, does not glow
 - (E) 4.5 V, glows fully
 - (F) 3.0 V, glows fully

4. Work out $2\frac{1}{2} + 1\frac{3}{4} - \frac{3}{8}$ and give the answer as a mixed number. [\[Fractions & Decimals\]](#)
 - (A) $4\frac{7}{8}$
 - (B) $4\frac{1}{8}$
 - (C) $4\frac{1}{4}$
 - (D) $3\frac{3}{4}$
 - (E) $3\frac{7}{8}$
 - (F) $3\frac{1}{8}$
 - (G) $3\frac{5}{8}$

5. A child eats a 240 g chapati. Three-eighths of it is starch, and $\frac{5}{6}$ of that starch is digested by the time the food leaves the stomach. How many grams of starch are still undigested? [\[Nutrition in Animals\]](#)

- (A) 45 g
- (B) 30 g
- (C) 18 g
- (D) 75 g
- (E) 90 g
- (F) 150 g
- (G) 15 g

6. A number is multiplied by 4 and then 7 is subtracted, giving 25. The same number is then tripled and 5 is added. What is this second result? [\[Simple Equations\]](#)

- (A) 8
- (B) 37
- (C) 32
- (D) 19
- (E) 26
- (F) 29

7. A heater coil turns 60 joules of electrical energy into heat every second. How many joules of heat does it give out if it runs for 2.5 minutes? [\[Electric Current & its Effects\]](#)

- (A) 60 J
- (B) 150 J
- (C) 3600 J
- (D) 15000 J
- (E) 900 J
- (F) 9000 J

8. A recipe needs 0.75 kg of flour to make 12 biscuits. How many grams of flour are needed for 20 biscuits at the same rate? [\[Fractions & Decimals\]](#)

- (A) 1250 g
- (B) 625 g
- (C) 450 g
- (D) 1200 g
- (E) 750 g
- (F) 1500 g
- (G) 900 g

9. In the small intestine, tiny villi absorb glucose. Suppose 5,000,000 villi each absorb 4 micrograms of glucose per minute. How many milligrams of glucose do they absorb together in one minute? (1 mg = 1000 micrograms) [\[Nutrition in Animals\]](#)

- (A) 5000 mg
- (B) 2 mg
- (C) 40000 mg
- (D) 20000 mg
- (E) 20 mg
- (F) 2000 mg
- (G) 200000 mg

10. The sum of three consecutive even numbers is 84. What is the largest of the three? [\[Simple Equations\]](#)

- (A) 24
- (B) 27
- (C) 30
- (D) 28
- (E) 32
- (F) 29
- (G) 26

11. In a series circuit, three identical bulbs share a 9 V cell's voltage equally. Two more identical bulbs are added in series. What voltage does each bulb get now? [\[Electric Current & its Effects\]](#)

- (A) 1.5 V
- (B) 1.8 V
- (C) 2 V
- (D) 2.25 V
- (E) 3 V
- (F) 9 V

12. Multiply 0.06 by 0.5, then divide the result by 0.003. What do you get? [\[Fractions & Decimals\]](#)

- (A) 100
- (B) 1000
- (C) 1
- (D) 0.01
- (E) 10
- (F) 0.1

13. A python swallows its prey and digests it slowly. It digests $\frac{1}{4}$ of the prey's mass on the first day, then $\frac{1}{3}$ of whatever is left on the second day. What fraction of the original prey remains after two days? [\[Nutrition in Animals\]](#)

- (A) $\frac{5}{12}$
- (B) $\frac{7}{12}$
- (C) $\frac{2}{7}$
- (D) $\frac{1}{4}$
- (E) $\frac{3}{8}$
- (F) $\frac{1}{2}$
- (G) $\frac{1}{3}$

14. In a circuit kit, the number of wires is 5 more than twice the number of bulbs. If there are 29 wires, how many bulbs are there? [\[Simple Equations\]](#)

- (A) 17
- (B) 10
- (C) 24
- (D) 29
- (E) 14
- (F) 12
- (G) 7

- 15.** A fuse melts if the current is above 5 A. Four appliances, each drawing 1.2 A, run together on it. Does the fuse blow, and by how much is the total above or below the limit? [\[Electric Current & its Effects\]](#)
- (A) Does not blow; 4.8 A below the limit
 - (B) Blows; exactly at the limit
 - (C) Does not blow; 1.2 A below the limit
 - (D) Blows; 0.2 A above the limit
 - (E) Blows; 4.8 A above the limit
 - (F) Does not blow; 0.8 A below the limit
 - (G) Does not blow; 0.2 A below the limit
- 16.** A tank is $\frac{4}{5}$ full. In the morning $\frac{1}{3}$ of the water present is used, and at night an amount equal to $\frac{1}{4}$ of the full tank is used. What fraction of the full tank remains? [\[Fractions & Decimals\]](#)
- (A) $\frac{7}{12}$
 - (B) $\frac{17}{60}$
 - (C) $\frac{11}{60}$
 - (D) $\frac{23}{60}$
 - (E) $\frac{1}{4}$
 - (F) $\frac{13}{60}$
- 17.** The human food canal is about 9 m long. The small intestine is about $\frac{2}{3}$ of the total length and the large intestine about $\frac{1}{6}$ of the total. How much longer is the small intestine than the large intestine? [\[Nutrition in Animals\]](#)
- (A) 7.5 m
 - (B) 1.5 m
 - (C) 3 m
 - (D) 6 m
 - (E) 4.5 m
 - (F) 4 m
- 18.** When a number is divided by 6 and 4 is added, the result equals one-third of the number. Find the number. [\[Simple Equations\]](#)
- (A) 48
 - (B) 12
 - (C) 8
 - (D) 6
 - (E) 36
 - (F) 18
 - (G) 24
- 19.** An electromagnet's lifting power grows in step with the number of wire turns. With 50 turns it lifts 20 iron pins. How many pins will 130 turns lift? [\[Electric Current & its Effects\]](#)
- (A) 50 pins
 - (B) 52 pins
 - (C) 100 pins
 - (D) 65 pins
 - (E) 40 pins
 - (F) 130 pins
 - (G) 26 pins

- 20.** The number 3.6 is first increased by 25%, and the result is then decreased by 25%. What is the final number? [\[Fractions & Decimals\]](#)
- (A) 3.375
 - (B) 3.4
 - (C) 4.05
 - (D) 2.7
 - (E) 4.5
 - (F) 3.6
 - (G) 3.5
- 21.** A goat chews each mouthful of cud 60 times and brings up 35 mouthfuls every hour. How many chews does it make in 2 hours and 30 minutes? [\[Nutrition in Animals\]](#)
- (A) 2100
 - (B) 4200
 - (C) 5250
 - (D) 3150
 - (E) 6300
 - (F) 5260
- 22.** Twice a number decreased by 9 equals the number increased by 6. What is three times the number? [\[Simple Equations\]](#)
- (A) 30
 - (B) 5
 - (C) 42
 - (D) 15
 - (E) 45
 - (F) 24
 - (G) 48
- 23.** A battery can run a certain bulb for 8 hours. It is swapped for a bulb that uses current 1.25 times as fast. How long will the same battery now last? [\[Electric Current & its Effects\]](#)
- (A) 6 hours 24 minutes
 - (B) 6 hours 40 minutes
 - (C) 6 hours 4 minutes
 - (D) 8 hours
 - (E) 7 hours
 - (F) 10 hours
- 24.** Among the values 0.41, $\frac{2}{5}$, 0.39, and $\frac{3}{8}$, which is the largest? [\[Fractions & Decimals\]](#)
- (A) 0.40
 - (B) 0.39
 - (C) $\frac{2}{5}$
 - (D) 0.04
 - (E) $\frac{3}{8}$
 - (F) they are all equal
 - (G) 0.41

- 25.** Bile from the liver helps digest fat. A meal has 48 g of fat. Bile emulsifies $\frac{3}{4}$ of it, and enzymes then break down $\frac{5}{6}$ of the emulsified fat. How many grams of fat do the enzymes break down? [\[Nutrition in Animals\]](#)
- (A) 6 g
 - (B) 12 g
 - (C) 45 g
 - (D) 36 g
 - (E) 40 g
 - (F) 30 g
- 26.** A rope is cut into two pieces so that the longer piece is 3 times the shorter piece. If the whole rope is 2.4 m long, how long is the shorter piece in centimetres? [\[Simple Equations\]](#)
- (A) 60 cm
 - (B) 120 cm
 - (C) 90 cm
 - (D) 180 cm
 - (E) 80 cm
 - (F) 40 cm
- 27.** A coil's heat output per second is proportional to the square of the current through it. At 2 A it gives out 12 J each second. How much heat per second does it give at 4 A? [\[Electric Current & its Effects\]](#)
- (A) 24 J/s
 - (B) 16 J/s
 - (C) 48 J/s
 - (D) 36 J/s
 - (E) 12 J/s
 - (F) 96 J/s
 - (G) 60 J/s
- 28.** What is $\frac{5}{8}$ of $\frac{2}{3}$ of 720? [\[Fractions & Decimals\]](#)
- (A) 360
 - (B) 480
 - (C) 240
 - (D) 600
 - (E) 375
 - (F) 450
 - (G) 300
- 29.** An Amoeba captures food with pseudopodia and forms one food vacuole every 9 minutes. How many food vacuoles can four Amoebae form together in 1 hour and 30 minutes? [\[Nutrition in Animals\]](#)
- (A) 36
 - (B) 10
 - (C) 40
 - (D) 20
 - (E) 80
 - (F) 30
 - (G) 45

30. A father is now 4 times as old as his son. In 5 years, the father will be 3 times as old as the son. How old is the son now? [\[Simple Equations\]](#)

- (A) 15
- (B) 40
- (C) 10
- (D) 12
- (E) 5
- (F) 20
- (G) 8

31. An electric bell's electromagnet makes the hammer vibrate 25 times each second. How many times does it vibrate in 3.6 minutes? [\[Electric Current & its Effects\]](#)

- (A) 540
- (B) 5400
- (C) 3600
- (D) 90
- (E) 216
- (F) 1500
- (G) 9000

32. A 1.6 m ribbon is cut into pieces each 0.125 m long. How many full pieces are obtained, and how much ribbon is left over? [\[Fractions & Decimals\]](#)

- (A) 12 pieces, 0.8 m left
- (B) 13 pieces, 0 m left
- (C) 11 pieces, 0.225 m left
- (D) 10 pieces, 0.35 m left
- (E) 12 pieces, 0.08 m left
- (F) 12 pieces, 0 m left
- (G) 12 pieces, 0.1 m left

33. An adult has 32 teeth. One-eighth of them are canines and one-half are molars. The remaining teeth are incisors and premolars in the ratio 1 : 2. How many premolars are there? [\[Nutrition in Animals\]](#)

- (A) 4
- (B) 2
- (C) 6
- (D) 12
- (E) 10
- (F) 16
- (G) 8

34. Solve for x: $(x/2) - (x/5) = 9$. [\[Simple Equations\]](#)

- (A) 15
- (B) 60
- (C) 45
- (D) 3
- (E) 30
- (F) 90
- (G) 18

- 35.** Two 1.5 V cells are joined so that their negative ends face each other, opposing one another. What is the net voltage across the pair? [\[Electric Current & its Effects\]](#)
- (A) 2.25 V
 - (B) 0 V
 - (C) 0.5 V
 - (D) 0.75 V
 - (E) 4.5 V
 - (F) 1.5 V
 - (G) 3 V
- 36.** Evaluate (0.2) cubed plus (0.3) squared. [\[Fractions & Decimals\]](#)
- (A) 0.98
 - (B) 0.13
 - (C) 0.158
 - (D) 0.18
 - (E) 0.098
 - (F) 0.107
 - (G) 0.0098
- 37.** A leech can hold blood equal to 5 times its own body mass. A leech of mass 2.4 g has already sucked 7.2 g of blood. How many more grams can it still hold? [\[Nutrition in Animals\]](#)
- (A) 19.2 g
 - (B) 7.2 g
 - (C) 4.8 g
 - (D) 5.6 g
 - (E) 12 g
 - (F) 9.6 g
- 38.** A shopkeeper sells pens at Rs 8 each and notebooks at Rs 15 each. He sells 5 more notebooks than pens and earns Rs 213 in all. How many pens did he sell? [\[Simple Equations\]](#)
- (A) 6
 - (B) 8
 - (C) 7
 - (D) 4
 - (E) 5
 - (F) 9
- 39.** A wire's resistance doubles every time its length doubles, and current halves whenever resistance doubles. A 2 m wire carries 6 A. The wire is replaced by an 8 m wire at the same voltage. What current flows now? [\[Electric Current & its Effects\]](#)
- (A) 3 A
 - (B) 1.5 A
 - (C) 6 A
 - (D) 12 A
 - (E) 2 A
 - (F) 0.75 A
 - (G) 24 A

40. A car covers 14.4 km on 1.2 litres of petrol. How far can it go on 5.5 litres at the same rate? [\[Fractions & Decimals\]](#)

- (A) 12 km
- (B) 55 km
- (C) 66 km
- (D) 72 km
- (E) 79.2 km
- (F) 60 km

41. A student tests 6 food samples with iodine; $\frac{2}{3}$ of them contain starch. Of the starch samples, half also contain protein. How many samples have starch but NOT protein? [\[Nutrition in Animals\]](#)

- (A) 5
- (B) 4
- (C) 0
- (D) 3
- (E) 2
- (F) 6
- (G) 1

42. The perimeter of a rectangle is 54 cm. Its length is 3 cm more than twice its breadth. Find the breadth. [\[Simple Equations\]](#)

- (A) 27 cm
- (B) 13 cm
- (C) 8 cm
- (D) 11 cm
- (E) 9 cm
- (F) 19 cm
- (G) 6 cm

43. An LED strip has 18 bulbs, each drawing 0.04 A, split equally into 3 parallel rows. What total current does the strip draw from the source? [\[Electric Current & its Effects\]](#)

- (A) 0.04 A
- (B) 7.2 A
- (C) 0.72 mA
- (D) 0.36 A
- (E) 0.72 A
- (F) 0.24 A

44. Simplify $(\frac{3}{4} \text{ divided by } \frac{2}{3})$ multiplied by $(\frac{5}{6} - \frac{1}{3})$. [\[Fractions & Decimals\]](#)

- (A) $\frac{3}{4}$
- (B) $\frac{9}{32}$
- (C) $\frac{3}{8}$
- (D) $\frac{5}{16}$
- (E) $\frac{9}{16}$
- (F) $\frac{1}{2}$
- (G) $\frac{9}{8}$

45. A cow eats 18 kg of grass a day. It brings back $\frac{5}{6}$ of it as cud and rechews $\frac{9}{10}$ of that cud. How many kilograms of grass get rechewed? [\[Nutrition in Animals\]](#)

- (A) 9 kg
- (B) 16.2 kg
- (C) 12 kg
- (D) 13.5 kg
- (E) 1.5 kg
- (F) 27 kg
- (G) 15 kg

46. Five times a number is as much above 60 as the number itself is below 60. Find the number. [\[Simple Equations\]](#)

- (A) 15
- (B) 40
- (C) 12
- (D) 30
- (E) 24
- (F) 10
- (G) 20

47. A circuit's switch is closed (ON) for $\frac{3}{5}$ of every minute and open (OFF) the rest of the time. Over one hour, for how many minutes is the bulb lit? [\[Electric Current & its Effects\]](#)

- (A) 30 minutes
- (B) 60 minutes
- (C) 20 minutes
- (D) 12 minutes
- (E) 24 minutes
- (F) 45 minutes
- (G) 36 minutes

48. The decimal 0.875 is written as a fraction in its lowest terms. What is the sum of its numerator and denominator? [\[Fractions & Decimals\]](#)

- (A) 78
- (B) 15
- (C) 1875
- (D) 875
- (E) 23
- (F) 16
- (G) 9

49. A salivary gland makes 1.5 litres of saliva in a day. If $\frac{2}{5}$ of it is used for breakfast-related digestion, how many millilitres of saliva is that? (1 litre = 1000 mL) [\[Nutrition in Animals\]](#)

- (A) 500 mL
- (B) 900 mL
- (C) 600 mL
- (D) 375 mL
- (E) 60 mL
- (F) 3750 mL
- (G) 1200 mL

50. A two-digit number has digits that add up to 11. The tens digit is 3 more than the units digit. What is the number? [\[Simple Equations\]](#)

- (A) 92
- (B) 65
- (C) 56
- (D) 74
- (E) 83
- (F) 47
- (G) 38

51. A compass needle near a current-carrying wire turns 8 degrees for every 0.5 A flowing. If it turns by 56 degrees, what current is in the wire? [\[Electric Current & its Effects\]](#)

- (A) 28 A
- (B) 14 A
- (C) 3.5 A
- (D) 7 A
- (E) 3 A
- (F) 0.5 A
- (G) 4 A

52. A wall is 2.4 m wide. Tiles 0.16 m wide are laid with a 0.04 m gap left at each end (two gaps). How many whole tiles fit across the wall? [\[Fractions & Decimals\]](#)

- (A) 13 tiles
- (B) 16 tiles
- (C) 15 tiles
- (D) 10 tiles
- (E) 12 tiles
- (F) 14 tiles

53. Of 40 organisms studied, $\frac{3}{5}$ are heterotrophs and the rest are autotrophs. Among the heterotrophs, $\frac{1}{4}$ are parasites. How many are heterotrophs that are NOT parasites? [\[Nutrition in Animals\]](#)

- (A) 12
- (B) 6
- (C) 16
- (D) 18
- (E) 10
- (F) 30

54. Solve for x: $7 - 2(x - 3) = 3x - 7$. [\[Simple Equations\]](#)

- (A) 5
- (B) 2
- (C) 4
- (D) 3
- (E) 1
- (F) 6

55. A rechargeable cell takes 90 minutes to charge fully, then runs a toy for 75 minutes before going flat. In a 5.5-hour session a child fully charges, plays, recharges, and plays again. How many minutes were spent NOT playing (charging)? [\[Electric Current & its Effects\]](#)

- (A) 75 minutes
- (B) 165 minutes
- (C) 120 minutes
- (D) 330 minutes
- (E) 180 minutes
- (F) 90 minutes
- (G) 150 minutes

56. Find the value of $7 - 0.07 - 0.7 - 0.007$. [\[Fractions & Decimals\]](#)

- (A) 6.223
- (B) 6.293
- (C) 6.217
- (D) 6.23
- (E) 6.923
- (F) 6.77

57. A stomach churns food for 4 hours, releasing 0.6 litres of gastric juice each hour. If $\frac{1}{3}$ of all the juice released is hydrochloric acid, how many litres of acid is that? [\[Nutrition in Animals\]](#)

- (A) 0.4 L
- (B) 2.4 L
- (C) 0.6 L
- (D) 0.8 L
- (E) 1.2 L
- (F) 0.2 L
- (G) 1.8 L

58. The three angles of a triangle are in the ratio 2 : 3 : 4. Find the largest angle. [\[Simple Equations\]](#)

- (A) 72 deg
- (B) 100 deg
- (C) 80 deg
- (D) 60 deg
- (E) 45 deg
- (F) 90 deg
- (G) 40 deg

59. Three resistors in series have values 4, 6, and 5 ohms. A fourth resistor is added so that the total becomes 3 times the original total. What is the value of the fourth resistor? [\[Electric Current & its Effects\]](#)

- (A) 3 ohms
- (B) 30 ohms
- (C) 20 ohms
- (D) 15 ohms
- (E) 10 ohms
- (F) 60 ohms

- 60.** Two-fifths of a class scored above 80%. Of the rest, $\frac{3}{4}$ scored between 60% and 80%. If the class has 40 students, how many scored below 60%? [\[Fractions & Decimals\]](#)
- (A) 10
 - (B) 24
 - (C) 12
 - (D) 8
 - (E) 18
 - (F) 6
- 61.** The small intestine absorbs 95% of a meal's nutrients. A second absorption stage then recovers half of whatever the small intestine missed. A meal has 80 g of absorbable nutrients. How many grams are absorbed in all? [\[Nutrition in Animals\]](#)
- (A) 77 g
 - (B) 78 g
 - (C) 74 g
 - (D) 76 g
 - (E) 79 g
 - (F) 4 g
 - (G) 80 g
- 62.** A taxi's fare in rupees is $F = 25 + 12k$, where k is the distance in kilometres. A passenger paid Rs 169. How many kilometres was the trip? [\[Simple Equations\]](#)
- (A) 144 km
 - (B) 7 km
 - (C) 169 km
 - (D) 14 km
 - (E) 10 km
 - (F) 12 km
 - (G) 13 km
- 63.** When a circuit is switched on, the current rises steadily from 0 to 2 A over 8 seconds. At what time is the current 1.25 A? [\[Electric Current & its Effects\]](#)
- (A) 2.5 seconds
 - (B) 3 seconds
 - (C) 10 seconds
 - (D) 1.25 seconds
 - (E) 5 seconds
 - (F) 4 seconds
- 64.** 0.6 of a number is 4.5 more than 0.4 of the same number. Find the number. [\[Fractions & Decimals\]](#)
- (A) 45
 - (B) 2.25
 - (C) 22.5
 - (D) 9
 - (E) 18
 - (F) 11.25
 - (G) 4.5

- 65.** A person eats 1.6 kg of food in a day. The body keeps $\frac{7}{8}$ of it and egests the rest. How many grams are egested? [\[Nutrition in Animals\]](#)
- (A) 1600 g
 - (B) 200 g
 - (C) 100 g
 - (D) 400 g
 - (E) 1400 g
 - (F) 800 g
- 66.** One-third of a number plus one-fourth of the same number equals 21. Find the number. [\[Simple Equations\]](#)
- (A) 18
 - (B) 36
 - (C) 7
 - (D) 252
 - (E) 42
 - (F) 12
 - (G) 28
- 67.** An electric kettle's coil gives out 250 J of heat each second. Boiling one cup of water needs 30,000 J. How many minutes does that take? [\[Electric Current & its Effects\]](#)
- (A) 0.5 minutes
 - (B) 120 minutes
 - (C) 2.0 minutes
 - (D) 3.0 minutes
 - (E) 2.4 minutes
 - (F) 12 minutes
- 68.** A jug holds 1.25 litres and a glass holds 0.15 litres. After filling 7 glasses from a full jug, how many litres are left in the jug? [\[Fractions & Decimals\]](#)
- (A) 0.05 L
 - (B) 1.1 L
 - (C) 0.25 L
 - (D) 0.15 L
 - (E) 1.05 L
 - (F) 0.3 L
 - (G) 0.2 L
- 69.** A Hydra has 8 tentacles, and each tentacle captures 3 water-fleas every 5 minutes. How many water-fleas can the Hydra catch in 40 minutes? [\[Nutrition in Animals\]](#)
- (A) 384
 - (B) 64
 - (C) 96
 - (D) 120
 - (E) 192
 - (F) 24

- 70.** A father shares Rs 1200 among three children. The second gets twice the first, and the third gets Rs 100 more than the second. How much does the first child get? [\[Simple Equations\]](#)
- (A) Rs 540
 - (B) Rs 240
 - (C) Rs 360
 - (D) Rs 200
 - (E) Rs 300
 - (F) Rs 220
- 71.** In a parallel circuit, each of 4 equal branches carries 0.6 A. One branch is switched off. By what fraction does the total current from the cell drop? [\[Electric Current & its Effects\]](#)
- (A) $\frac{2}{3}$
 - (B) $\frac{1}{3}$
 - (C) $\frac{1}{6}$
 - (D) $\frac{3}{4}$
 - (E) $\frac{1}{4}$
 - (F) 0.6
 - (G) $\frac{1}{2}$
- 72.** Which is greater, $\frac{5}{6}$ or 0.8, and by how much? Give the difference as a fraction. [\[Fractions & Decimals\]](#)
- (A) $\frac{5}{6}$ is greater by $\frac{1}{15}$
 - (B) $\frac{5}{6}$ is greater by $\frac{1}{6}$
 - (C) 0.8 is greater by $\frac{1}{30}$
 - (D) $\frac{5}{6}$ is greater by $\frac{1}{30}$
 - (E) $\frac{5}{6}$ is greater by $\frac{1}{5}$
 - (F) 0.8 is greater by $\frac{1}{6}$
- 73.** The liver makes 0.75 L of bile a day, which the gall bladder concentrates down to $\frac{1}{5}$ of its volume. What volume of concentrated bile does the gall bladder hold? (1 litre = 1000 mL) [\[Nutrition in Animals\]](#)
- (A) 600 mL
 - (B) 250 mL
 - (C) 150 mL
 - (D) 75 mL
 - (E) 300 mL
 - (F) 375 mL
 - (G) 900 mL
- 74.** If 4 is added to a number and the sum is multiplied by 3, the result is 5 more than 4 times the number. Find the number. [\[Simple Equations\]](#)
- (A) 7
 - (B) 5
 - (C) 9
 - (D) -1
 - (E) 3
 - (F) 12
 - (G) 11

75. A circuit tester beeps once each second while current flows and stays silent when the circuit is broken. During a 4-minute test the circuit was broken for 35 seconds in total. How many times did the tester beep?

[Electric Current & its Effects]

- (A) 200
- (B) 175
- (C) 35
- (D) 245
- (E) 205
- (F) 240
- (G) 215

76. Multiply the mixed number $3\frac{1}{3}$ by 2.4 and give the answer as a decimal. [Fractions & Decimals]

- (A) 7.92
- (B) 8.2
- (C) 8.4
- (D) 6.72
- (E) 9.6
- (F) 7.2
- (G) 8

77. Among 60 animals, $\frac{1}{4}$ are filter feeders, $\frac{2}{5}$ are grazers, and the rest are predators. How many more grazers are there than filter feeders? [Nutrition in Animals]

- (A) 6
- (B) 39
- (C) 15
- (D) 9
- (E) 24
- (F) 12

78. A science kit holds only resistors and switches, 40 pieces in all, with 8 fewer switches than resistors. How many resistors are there? [Simple Equations]

- (A) 12
- (B) 20
- (C) 32
- (D) 24
- (E) 28
- (F) 18

79. Doubling the current doubles an electromagnet's lifting force. At 3 A it lifts 150 g. What current is needed to lift a 400 g load? [Electric Current & its Effects]

- (A) 3 A
- (B) 6 A
- (C) 9 A
- (D) 16 A
- (E) 8 A
- (F) 12 A

- 80.** A number line is marked at every 0.25. Starting at 1.5 and moving 11 steps to the right, where do you land? [\[Fractions & Decimals\]](#)
- (A) 3.25
 - (B) 2.75
 - (C) 12.5
 - (D) 4.5
 - (E) 4.0
 - (F) 4.25
- 81.** An adult has about 10,500 taste buds, and each is replaced every 2 weeks. About how many taste buds are replaced per day on average? (1 week = 7 days) [\[Nutrition in Animals\]](#)
- (A) 525
 - (B) 1050
 - (C) 7500
 - (D) 5250
 - (E) 1500
 - (F) 750
 - (G) 375
- 82.** Solve for y : $(2y + 3)/5 = (y - 1)/2$. [\[Simple Equations\]](#)
- (A) -11
 - (B) 6
 - (C) 11
 - (D) 13
 - (E) 9
 - (F) 5
 - (G) 1
- 83.** A solar lamp charges at 12% per hour in sunlight and discharges at 20% per hour when lit. It starts empty, charges for 5 hours, then is switched on. How many hours of light does it give before going flat? [\[Electric Current & its Effects\]](#)
- (A) 6 hours
 - (B) 5 hours
 - (C) 12 hours
 - (D) 4 hours
 - (E) 3 hours
 - (F) 2.4 hours
- 84.** Evaluate 0.45 divided by 0.9, then add 0.6 multiplied by 0.5. [\[Fractions & Decimals\]](#)
- (A) 0.5
 - (B) 0.8
 - (C) 0.35
 - (D) 0.3
 - (E) 0.08
 - (F) 1.05
 - (G) 0.55

85. A Paramecium beats its cilia 1,200 times a minute to sweep food into its mouth groove. How many times does it beat its cilia in $\frac{3}{4}$ of an hour? [\[Nutrition in Animals\]](#)

- (A) 5400
- (B) 108000
- (C) 45000
- (D) 54000
- (E) 36000
- (F) 900

86. A bag of marbles is shared so that A gets $\frac{1}{2}$, B gets $\frac{1}{3}$, and C gets the remaining 8 marbles. How many marbles are there in total? [\[Simple Equations\]](#)

- (A) 24
- (B) 16
- (C) 8
- (D) 12
- (E) 30
- (F) 48
- (G) 40

87. A circuit diagram uses 2 cells, 3 bulbs, 1 switch, and connecting wires. Each cell is drawn with 2 lines, each bulb as 1 circle, and the switch as 1 gap. How many circles are drawn in the whole diagram? [\[Electric Current & its Effects\]](#)

- (A) 4
- (B) 3
- (C) 9
- (D) 2
- (E) 1
- (F) 6

88. Three friends share a 2.7 kg cake. The first takes $\frac{1}{3}$ of it, the second takes 0.4 of the whole, and the third takes the rest. How many grams does the third friend get? [\[Fractions & Decimals\]](#)

- (A) 720 g
- (B) 900 g
- (C) 630 g
- (D) 810 g
- (E) 1080 g
- (F) 540 g

89. A 2000 kcal meal plan gives $\frac{1}{2}$ of its energy from carbohydrates, $\frac{3}{10}$ from fats, and the rest from protein. How many kilocalories come from protein? [\[Nutrition in Animals\]](#)

- (A) 1600 kcal
- (B) 500 kcal
- (C) 600 kcal
- (D) 200 kcal
- (E) 800 kcal
- (F) 1000 kcal
- (G) 400 kcal

90. Three times a number, decreased by 7, equals the number increased by 13. What is half of the number? [\[Simple Equations\]](#)

- (A) 5
- (B) 20
- (C) 7
- (D) 10
- (E) 13
- (F) 2.5
- (G) 15

91. A coil warms 200 g of water. Raising water by 1 degree C needs about 4.2 J for each gram. How many joules are needed to raise 200 g of water by 15 degrees C? [\[Electric Current & its Effects\]](#)

- (A) 12600 J
- (B) 6300 J
- (C) 840 J
- (D) 1260 J
- (E) 12060 J
- (F) 3000 J
- (G) 25200 J

92. A pen costs Rs 12.50 on its own. A box of 24 pens is sold for Rs 270. How much is saved per pen by buying the box? [\[Fractions & Decimals\]](#)

- (A) Rs 2.50
- (B) Rs 30
- (C) Rs 11.25
- (D) Rs 0.75
- (E) Rs 1.25
- (F) Rs 1.20
- (G) Rs 1.50

93. A tapeworm absorbs 2.5 g of digested food a day directly through its skin. If it lives in the gut for 8 months, how many kilograms of food does it absorb? (1 month = 30 days, 1 kg = 1000 g) [\[Nutrition in Animals\]](#)

- (A) 6 kg
- (B) 1.2 kg
- (C) 60 kg
- (D) 0.72 kg
- (E) 0.06 kg
- (F) 0.6 kg
- (G) 0.3 kg

94. The cost of 3 erasers and 2 sharpeners is Rs 48. A sharpener costs Rs 4 more than an eraser. Find the cost of one sharpener. [\[Simple Equations\]](#)

- (A) Rs 14
- (B) Rs 10
- (C) Rs 4
- (D) Rs 9
- (E) Rs 8
- (F) Rs 12

- 95.** A model train's motor draws 0.8 A. The track's battery can supply a total of 4 A. How many such trains can run at the same time without going over the supply? [\[Electric Current & its Effects\]](#)
- (A) 4 trains
 - (B) 5 trains
 - (C) 8 trains
 - (D) 10 trains
 - (E) 6 trains
 - (F) 3 trains
- 96.** Write $\frac{3}{8} + 0.6$ as a single decimal. [\[Fractions & Decimals\]](#)
- (A) 0.9
 - (B) 0.975
 - (C) 0.435
 - (D) 0.38
 - (E) 0.985
 - (F) 1.35
- 97.** Food takes about 4 hours in the stomach, then 6 hours in the small intestine, and the rest of a 24-hour journey in the large intestine. What fraction of the whole journey is spent in the large intestine? [\[Nutrition in Animals\]](#)
- (A) $\frac{1}{2}$
 - (B) $\frac{1}{3}$
 - (C) $\frac{5}{12}$
 - (D) $\frac{2}{3}$
 - (E) $\frac{10}{24}$
 - (F) $\frac{7}{12}$
- 98.** When the same number is added to both the top and bottom of the fraction $\frac{5}{7}$, the fraction becomes $\frac{3}{4}$. Find that number. [\[Simple Equations\]](#)
- (A) 4
 - (B) 1
 - (C) 3
 - (D) 6
 - (E) 2
 - (F) 5
 - (G) 7
- 99.** Bulb X uses electrical energy twice as fast as bulb Y. Working together for one minute they use 540 J in total. How many joules does bulb X use in that minute? [\[Electric Current & its Effects\]](#)
- (A) 300 J
 - (B) 120 J
 - (C) 405 J
 - (D) 270 J
 - (E) 360 J
 - (F) 180 J

100. A tank fills to 0.8 of its volume in 12 minutes at a steady rate. How many more minutes are needed to fill it completely? [\[Fractions & Decimals\]](#)

- (A) 3 minutes
- (B) 6 minutes
- (C) 15 minutes
- (D) 4 minutes
- (E) 2.4 minutes
- (F) 9 minutes
- (G) 12 minutes